



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)



Decision Tree

Dr. Nirmala Baloorkar

Assistant Professor

Department of Computer Engineering KJSSE

Gini Index

Gini index or Gini impurity measures the degree or probability of a particular variable being wrongly classified when it is randomly chosen.

But what is actually meant by 'impurity'?

If all the elements belong to a single class, then it can be called pure.

The degree of Gini index varies between 0 and 1, where 0 denotes that all elements belong to a certain class or if there exists only one class,

1 denotes that the elements are randomly distributed across various classes.

A Gini Index of 0.5 denotes equally distributed elements into some classes.

Formula for Gini Index

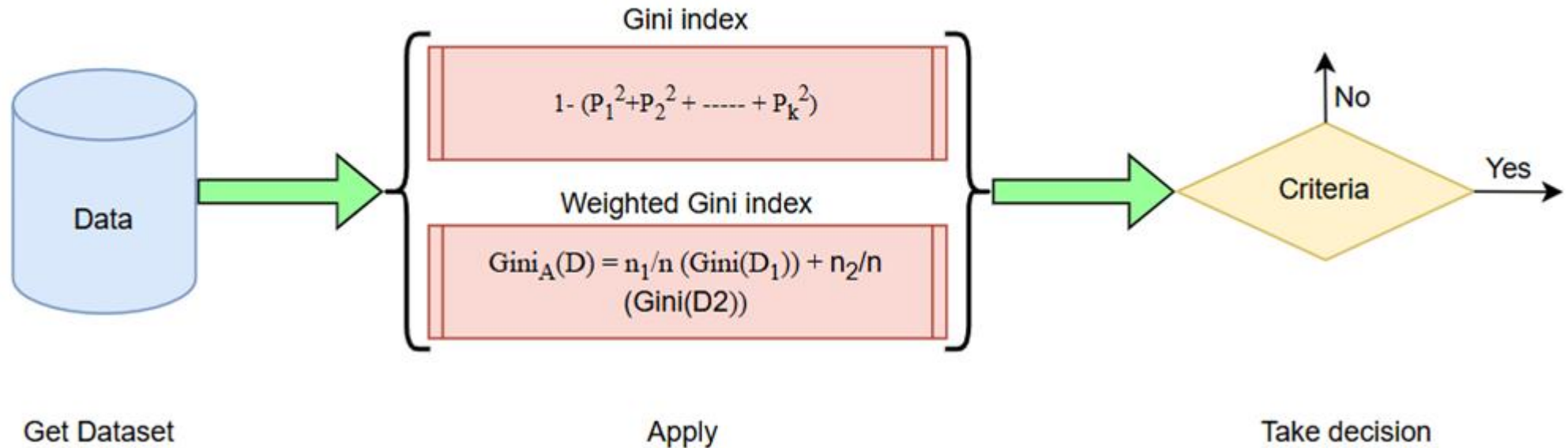
$$\text{Gini} = 1 - \sum_{i=1}^n (p_i)^2$$

where p_i is the probability of an object being classified to a particular class.

While building the decision tree, we would prefer choosing the attribute/feature with the least Gini index as the root node.

Decision Tree Using Gini Index

Construct decision Tree using Gini Index



Example

Past Trend	Open Interest	Trading Volume	Return
Positive	Low	High	Up
Negative	High	Low	Down
Positive	Low	High	Up
Positive	High	High	Up
Negative	Low	High	Down
Positive	Low	Low	Down
Negative	High	High	Down
Negative	Low	High	Down
Positive	Low	Low	Down
Positive	High	High	Up

Calculating the Gini Index for 'Past Trend'

P(Past Trend=Positive): 6/10

P(Past Trend=Negative): 4/10

If (Past Trend = Positive & Return = Up), probability = 4/6

If (Past Trend = Positive & Return = Down), probability = 2/6

Gini index = $1 - ((4/6)^2 + (2/6)^2) = 0.45$

If (Past Trend = Negative & Return = Up), probability = 0

If (Past Trend = Negative & Return = Down), probability = 4/4

Gini index = $1 - ((0)^2 + (4/4)^2) = 0$

Weighted sum of the Gini Indices can be calculated as follows:

Gini Index for Past Trend = $(6/10)0.45 + (4/10)0 = 0.27$

Past Trend	Open Interest	Trading Volume	Return
Positive	Low	High	Up
Negative	High	Low	Down
Positive	Low	High	Up
Positive	High	High	Up
Negative	Low	High	Down
Positive	Low	Low	Down
Negative	High	High	Down
Negative	Low	High	Down
Positive	Low	Low	Down
Positive	High	High	Up

calculation of Gini Index for Open Interest

P(Open Interest=High): 4/10

P(Open Interest=Low): 6/10

If (Open Interest = High & Return = Up), probability = 2/4

If (Open Interest = High & Return = Down), probability = 2/4

Gini index = $1 - ((2/4)^2 + (2/4)^2) = 0.5$

If (Open Interest = Low & Return = Up), probability = 2/6

If (Open Interest = Low & Return = Down), probability = 4/6

Gini index = $1 - ((2/6)^2 + (4/6)^2) = 0.45$

Weighted sum of the Gini Indices can be calculated as follows:

Gini Index for Open Interest = $(4/10)0.5 + (6/10)0.45 = 0.47$

Past Trend	Open Interest	Trading Volume	Return
Positive	Low	High	Up
Negative	High	Low	Down
Positive	Low	High	Up
Positive	High	High	Up
Negative	Low	High	Down
Positive	Low	Low	Down
Negative	High	High	Down
Negative	Low	High	Down
Positive	Low	Low	Down
Positive	High	High	Up

Calculation of Gini Index for Trading Volume

P(Trading Volume=High): 7/10

P(Trading Volume=Low): 3/10

If (Trading Volume = High & Return = Up), probability = 4/7

If (Trading Volume = High & Return = Down), probability = 3/7

Gini index = $1 - ((4/7)^2 + (3/7)^2) = 0.49$

If (Trading Volume = Low & Return = Up), probability = 0

If (Trading Volume = Low & Return = Down), probability = 3/3

Gini index = $1 - ((0)^2 + (1)^2) = 0$

Past Trend	Open Interest	Trading Volume	Return
Positive	Low	High	Up
Negative	High	Low	Down
Positive	Low	High	Up
Positive	High	High	Up
Negative	Low	High	Down
Positive	Low	Low	Down
Negative	High	High	Down
Negative	Low	High	Down
Positive	Low	Low	Down
Positive	High	High	Up

Weighted sum of the Gini Indices can be calculated as follows:

Gini Index for Trading Volume = $(7/10)0.49 + (3/10)0 = 0.34$

Gini Index

Attributes/Features	Gini Index
Past Trend	0.27
Open Interest	0.47
Trading Volume	0.34

We observe that '**Past Trend**' has the lowest Gini Index and hence it will be chosen as the root node for how decision tree works.

We will repeat the same procedure to determine the sub-nodes or branches of the decision tree.

Calculate the Gini Index for the 'Positive' branch of Past Trend

Past Trend	Open Interest	Trading Volume	Return
Positive	Low	High	Up
Positive	Low	High	Up
Positive	High	High	Up
Positive	Low	Low	Down
Positive	Low	Low	Down
Positive	High	High	Up

Calculation of Gini Index of Open Interest for Positive Past Trend

P(Open Interest=High): 2/6

P(Open Interest=Low): 4/6

If (Open Interest = High & Return = Up), probability = 2/2

If (Open Interest = High & Return = Down), probability = 0

Gini index = $1 - ((2/2)^2 + (0)^2) = 0$

If (Open Interest = Low & Return = Up), probability = 2/4

If (Open Interest = Low & Return = Down), probability = 2/4

Gini index = $1 - ((2/4)^2 + (2/4)^2) = 0.50$ Weighted sum of the Gini Indices can be calculated as follows:

Gini Index for Open Interest = $(2/6)0 + (4/6)0.50 = 0.33$

Past Trend	Open Interest	Trading Volume	Return
Positive	Low	High	Up
Positive	Low	High	Up
Positive	High	High	Up
Positive	Low	Low	Down
Positive	Low	Low	Down
Positive	High	High	Up

Calculation of Gini Index for Trading Volume Positive past trend

P(Trading Volume=High): 4/6

P(Trading Volume=Low): 2/6

If (Trading Volume = High & Return = Up), probability = 4/4

If (Trading Volume = High & Return = Down), probability = 0

Gini index = $1 - ((4/4)^2 + (0)^2) = 0$

If (Trading Volume = Low & Return = Up), probability = 0

If (Trading Volume = Low & Return = Down), probability = 2/2

Gini index = $1 - ((0)^2 + (2/2)^2) = 0$

Past Trend	Open Interest	Trading Volume	Return
Positive	Low	High	Up
Positive	Low	High	Up
Positive	High	High	Up
Positive	Low	Low	Down
Positive	Low	Low	Down
Positive	High	High	Up

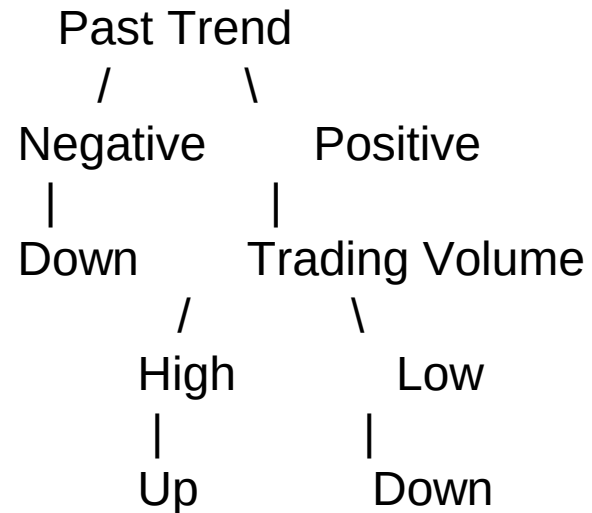
Weighted sum of the Gini Indices can be calculated as follows:

Gini Index for Trading Volume = $(4/6)0 + (2/6)0 = 0$

Gini Index

Attribute/Feature	Gini Index
Open Interest	0.33
Trading Volume	0

We will split the node further using the '**Trading Volume**' feature, as it has the minimum Gini index.



Key Differences in Decision Tree Splits

Criterion	Gini Impurity	Entropy
Computation	Faster (no logarithms)	Slower (requires log base 2)
Split Bias	Tends to favor larger, dominant classes	Produces more balanced splits
Sensitivity	Less sensitive to small probability changes	More sensitive to subtle differences
Best Use Case	Large, high-dimensional datasets	Balanced datasets where information gain matters
Library Defaults	Often default in CART (Classification and Regression Trees)	Common in ID3, C4.5, and theoretical ML research

Solve using Gini Index

CGPA	Inter active	Practical Knowledge	Comm Skills	Job Offer
≥ 9	Yes	Very good	Good	Yes
≥ 8	No	Good	Moderate	Yes
≥ 9	No	Average	Poor	No
< 8	No	Average	Good	No
≥ 8	Yes	Good	Moderate	Yes
≥ 9	Yes	Good	Moderate	Yes
< 8	Yes	Good	Poor	No
≥ 9	No	Very good	Good	Yes
≥ 8	Yes	Good	Good	Yes
≥ 8	Yes	Average	Good	Yes